

TEST REPORT

No. : XMIN2305000232CM01_EN

Date : 2023-05-17

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Sample Name : PP FLOOR FILM-SANDBLAST

Above information and sample(s) was/were submitted and confirmed by the client. SGS, however, assumes no responsibility to verify the accuracy, adequacy and completeness of the sample information provided by client.

SGS Ref. No. : CP23-020647, GZMR230501228801
Date of Receipt : 2023-05-10
Testing Period : 2023-05-10 ~ 2023-05-17
Test result(s) : For further details, please refer to the following page(s)
(Unless otherwise stated the results shown in this test report refer only to the sample(s) tested)

Signed for
SGS-CSTC Standards Technical
Services Co.,Ltd. Xiamen Branch.

Bryan

Bryan Hong
Authorized signatory



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Summary of Results:

No.	Test Item	Test Method	Result
1	Shod Ramp Test	DIN EN 16165:2021-12 Annex B	Ramp test value α_{shod} : 10° Slip resistance class: R10
2	Martindale Abrasion Test*	GB/T 21196.1-2007	Visible wear

Note: * test project/method was carried out by subcontractors.

Original Sample Photo:



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1. Test Item: Shod ramp test

Sample Description: Flooring, See photo

Test Method: DIN EN 16165:2021-12 Determination of slip resistance of pedestrian surfaces -

Methods of evaluation Annex B Shod ramp test

Test Condition:

Specimen: 1000mm×500mm, see photo

Testing surface: see photo

Testing direction: length direction

Test result:

Test item(s)	Test result(s)	Slip resistance class
Slip resistance - Shod ramp test	Ramp test value α_{shod} : 10°	R10

DIN EN 16165:2021-12 NA.2 Classification of the results by shod ramp test:

Test result α_{shod}	Slip resistance class
$6^\circ \leq \alpha_{shod} < 10^\circ$	R9
$10^\circ \leq \alpha_{shod} < 19^\circ$	R10
$19^\circ \leq \alpha_{shod} < 27^\circ$	R11
$27^\circ \leq \alpha_{shod} < 35^\circ$	R12
$35^\circ \leq \alpha_{shod}$	R13

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Test Property	Test procedure/requirements	Rating/Result
SLIP RESISTANCE	Prepare the test sample as the size of 100cm x 50cm, cleaning the surface before test. The temperature of the lab, shoes, lubricant and the decorative panels should be kept at $(23 \pm 5) ^\circ\text{C}$ Before test, apply a layer of lubricant on the surface of the decorative panels evenly with a brush, the density should be (200 ± 20) ml every square meter, the outer bottom of the shoes should also be covered with lubricant.  Illustration 1–bottom of the shoes for inspect Inspector should maintain upright posture and walk forward and backward on the decorative panels while watch below, stride width should reach half the length of the shoes. Start from the horizontality; Increase the angle of inclination of the panels at a angular velocity of about one degree every second. Inspector will linger at critical areas many times to determine the reliable walk limit inclination angle he or she can reached, repeat the above procedure three times and start from the horizontality every time. Before the second and the third time, reapply the lubricant on the surface as above with the brush. 1.Calibration of the test person 1. Each inspectors should walk on the every standard flooring for three times, then calculate the average angle respectively: ① $\alpha_{\text{KST-Ij}}$ ② $\alpha_{\text{KST-IIj}}$ ③ $\alpha_{\text{KST-IIIj}}$	$\alpha: 10^\circ$ Rating: R10

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Test Property	Test procedure/requirements	Rating/Result																														
	2. The difference value will be calculated: $\Delta\alpha_{ST-Ij}$, $\Delta\alpha_{ST-IIj}$, $\Delta\alpha_{ST-IIIj}$. If the difference value is out of range of CrD95, the inspector should be eliminated See below table 1 <table><tr><th colspan="3">Table 1</th></tr><tr><th colspan="3">Standard flooring</th></tr><tr><th>i</th><th>$\alpha_{S,i}$</th><th>CrD95</th></tr><tr><td>St-I</td><td>8.7°</td><td>3.0°</td></tr><tr><td>St-II</td><td>17.3°</td><td>3.0°</td></tr><tr><td>St-III A</td><td>27.3°</td><td>3.0°</td></tr></table> 2. Test for sample Two qualified inspector selected as above walk on the sample panels for three times respectively,, then calculate the mean value $\alpha_{0,1}$ and $\alpha_{0,2}$, then calculate the corrected value Dj according to below table 2.. <table><tr><th colspan="2">Table 2</th></tr><tr><th>Case</th><th>Corrected value Dj</th></tr><tr><td>$\alpha_{0,1} < \alpha_{K,St-I,1}$</td><td>$D_1 = \Delta\alpha_{St-I,1} \cdot \frac{1}{\sqrt{2}}$</td></tr><tr><td>$\alpha_{K,St-I,1} \leq \alpha_{0,1} < \alpha_{K,St-II,1}$</td><td>$D_1 = \left[\Delta\alpha_{St-I,1} + (\Delta\alpha_{St-II,1} - \Delta\alpha_{St-I,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-I,1}}{\alpha_{K,St-II,1} - \alpha_{K,St-I,1}} \right] \cdot \frac{1}{\sqrt{2}}$</td></tr><tr><td>$\alpha_{K,St-II,1} \leq \alpha_{0,1} < \alpha_{K,St-III A,1}$</td><td>$D_1 = \left[\Delta\alpha_{St-II,1} + (\Delta\alpha_{St-III A,1} - \Delta\alpha_{St-II,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-II,1}}{\alpha_{K,St-III A,1} - \alpha_{K,St-II,1}} \right] \cdot \frac{1}{\sqrt{2}}$</td></tr><tr><td>$\alpha_{K,St-III A,1} \leq \alpha_{0,1}$</td><td>$D_1 = \Delta\alpha_{St-III A,1} \cdot \frac{1}{\sqrt{2}}$</td></tr></table> The result for inspector j: $\alpha_j = \alpha_{0,j} + D_j$ The final result for the two inspectors: $\alpha = (\alpha_1 + \alpha_2) / 2$, on this basis and according to table 3, give a final rating of slip resistance.	Table 1			Standard flooring			i	$\alpha_{S,i}$	CrD95	St-I	8.7°	3.0°	St-II	17.3°	3.0°	St-III A	27.3°	3.0°	Table 2		Case	Corrected value Dj	$\alpha_{0,1} < \alpha_{K,St-I,1}$	$D_1 = \Delta\alpha_{St-I,1} \cdot \frac{1}{\sqrt{2}}$	$\alpha_{K,St-I,1} \leq \alpha_{0,1} < \alpha_{K,St-II,1}$	$D_1 = \left[\Delta\alpha_{St-I,1} + (\Delta\alpha_{St-II,1} - \Delta\alpha_{St-I,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-I,1}}{\alpha_{K,St-II,1} - \alpha_{K,St-I,1}} \right] \cdot \frac{1}{\sqrt{2}}$	$\alpha_{K,St-II,1} \leq \alpha_{0,1} < \alpha_{K,St-III A,1}$	$D_1 = \left[\Delta\alpha_{St-II,1} + (\Delta\alpha_{St-III A,1} - \Delta\alpha_{St-II,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-II,1}}{\alpha_{K,St-III A,1} - \alpha_{K,St-II,1}} \right] \cdot \frac{1}{\sqrt{2}}$	$\alpha_{K,St-III A,1} \leq \alpha_{0,1}$	$D_1 = \Delta\alpha_{St-III A,1} \cdot \frac{1}{\sqrt{2}}$	
Table 1																																
Standard flooring																																
i	$\alpha_{S,i}$	CrD95																														
St-I	8.7°	3.0°																														
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Case	Corrected value Dj																															
$\alpha_{0,1} < \alpha_{K,St-I,1}$	$D_1 = \Delta\alpha_{St-I,1} \cdot \frac{1}{\sqrt{2}}$																															
$\alpha_{K,St-I,1} \leq \alpha_{0,1} < \alpha_{K,St-II,1}$	$D_1 = \left[\Delta\alpha_{St-I,1} + (\Delta\alpha_{St-II,1} - \Delta\alpha_{St-I,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-I,1}}{\alpha_{K,St-II,1} - \alpha_{K,St-I,1}} \right] \cdot \frac{1}{\sqrt{2}}$																															
$\alpha_{K,St-II,1} \leq \alpha_{0,1} < \alpha_{K,St-III A,1}$	$D_1 = \left[\Delta\alpha_{St-II,1} + (\Delta\alpha_{St-III A,1} - \Delta\alpha_{St-II,1}) \cdot \frac{\alpha_{0,1} - \alpha_{K,St-II,1}}{\alpha_{K,St-III A,1} - \alpha_{K,St-II,1}} \right] \cdot \frac{1}{\sqrt{2}}$																															
$\alpha_{K,St-III A,1} \leq \alpha_{0,1}$	$D_1 = \Delta\alpha_{St-III A,1} \cdot \frac{1}{\sqrt{2}}$																															

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	Table3 –The relation between the corrected overall angle and the rating of the slip resistance <table><tr><th>α</th><th>Rating</th></tr><tr><td>$6^{\circ} \leq \alpha \leq 10^{\circ}$</td><td>R 9</td></tr><tr><td>$10^{\circ} \leq \alpha \leq 19^{\circ}$</td><td>R 10</td></tr><tr><td>$19^{\circ} \leq \alpha \leq 27^{\circ}$</td><td>R 11</td></tr><tr><td>$27^{\circ} \leq \alpha \leq 35^{\circ}$</td><td>R 12</td></tr><tr><td>$35^{\circ} \leq \alpha$</td><td>R 13</td></tr></table> Illustration 2 inspect device with safety mechanism (inclined plane)	α	Rating	$6^{\circ} \leq \alpha \leq 10^{\circ}$	R 9	$10^{\circ} \leq \alpha \leq 19^{\circ}$	R 10	$19^{\circ} \leq \alpha \leq 27^{\circ}$	R 11	$27^{\circ} \leq \alpha \leq 35^{\circ}$	R 12	$35^{\circ} \leq \alpha$	R 13	
α	Rating													
$6^{\circ} \leq \alpha \leq 10^{\circ}$	R 9													
$10^{\circ} \leq \alpha \leq 19^{\circ}$	R 10													
$19^{\circ} \leq \alpha \leq 27^{\circ}$	R 11													
$27^{\circ} \leq \alpha \leq 35^{\circ}$	R 12													
$35^{\circ} \leq \alpha$	R 13													

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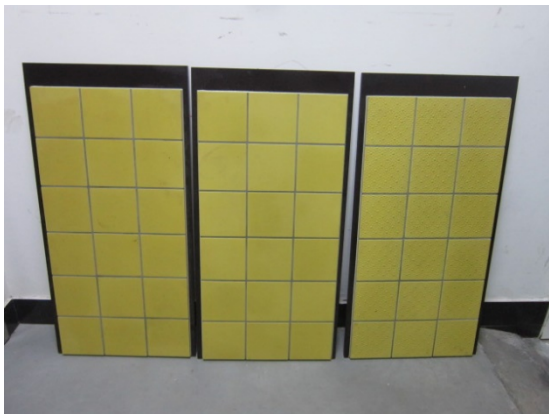
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Test Photo:

Standard floor



Test shoe



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2. Test Item: Martindale Abrasion Test

Sample Description: Flooring, See photo

Test Method: GB/T 21196.1-2007

Test Condition:

Rubbing finger: External diameter $\Phi 38\text{mm}$ (Contact area $\Phi 28.7\text{mm}$)

Rubbing material: No.600 waterproof abrasive paper

Speed: 47.5 rpm

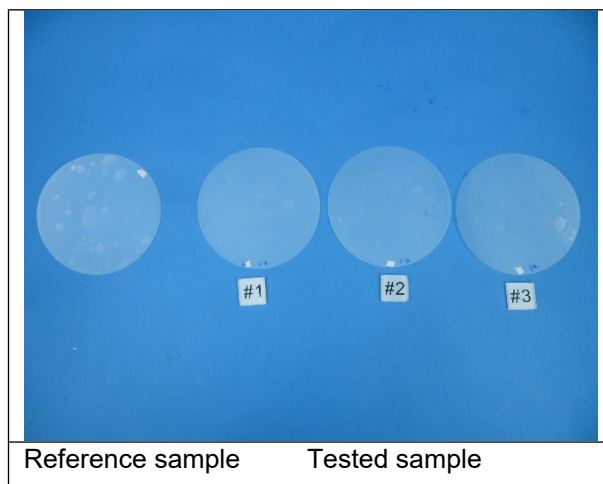
Load: 12kPa

Cycles:2000 cycles

Test Result:

Test Item	Result
Martindale Abrasion Test	Visible wear

Test Photo:



Reference sample Tested sample

*****End of report*****